

Domain: Values that input variable can take.

Example: $\cdot \{ (x, y, z) \mid \text{constraints} \}$
 $\cdot \mathbb{R}^3 \text{ except } \dots$

Type of constraints:

$$\cdot \sqrt{\quad} \Rightarrow \quad \geq 0.$$

$$\cdot \sin^{-1}(\quad), \cos^{-1}(\quad) \Rightarrow -1 \leq \quad \leq 1.$$

$$\cdot \tan(\quad) \Rightarrow \quad \neq \frac{\pi}{2} + k\pi \text{ for any integer } k.$$

$$\cdot \frac{1}{\quad} \Rightarrow \quad \neq 0$$

$$\cdot \ln(\quad) \Rightarrow \quad > 0.$$

2D
↓3D
↓

Level curve / Level surface

The set of points in the plane where a function $f(x, y)$ has a constant value $f(x, y) = c$ is called a level curve

The set of points (x, y, z) in space where a function of 3 independent variables has a constant value $f(x, y, z) = c$ is called of level surface.

To Find a level curve or surface at a point (x_0, y_0) or (x_0, y_0, z_0)

Plug in (x_0, y_0) or (x_0, y_0, z_0) to get C

$f(x, y) = C = f(x_0, y_0)$ or $f(x, y, z) = C = f(x_0, y_0, z_0)$.
expression of x & y ← constant expression of x, y & z ← constant.

Level curve vs. Contour Curve

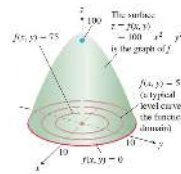


FIGURE 14.5 The graph and selected level curves of the function $f(x, y)$ in Example 3. The level curves lie in the xy -plane, which is the domain of the function $f(x, y)$.

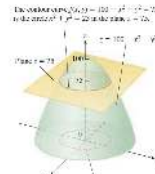
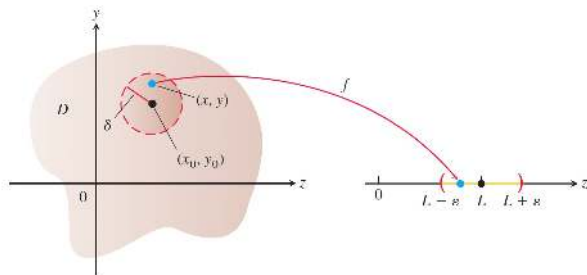


FIGURE 14.6 A plane $z = c$ parallel to the xy -plane intersecting a surface $z = f(x, y)$ produces a contour curve.

Limits



THEOREM 1—Properties of Limits of Functions of Two Variables

The following rules hold if L , M , and k are real numbers and

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L \quad \text{and} \quad \lim_{(x, y) \rightarrow (x_0, y_0)} g(x, y) = M.$$

1. *Sum Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} (f(x, y) + g(x, y)) = L + M$$

2. *Difference Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} (f(x, y) - g(x, y)) = L - M$$

3. *Constant Multiple Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} kf(x, y) = kL \quad (\text{any number } k)$$

4. *Product Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} (f(x, y) \cdot g(x, y)) = L \cdot M$$

5. *Quotient Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} \frac{f(x, y)}{g(x, y)} = \frac{L}{M}, \quad M \neq 0$$

6. *Power Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} [f(x, y)]^n = L^n, \quad n \text{ a positive integer}$$

7. *Root Rule:*

$$\lim_{(x, y) \rightarrow (x_0, y_0)} \sqrt[n]{f(x, y)} = \sqrt[n]{L} = L^{1/n},$$

n a positive integer, and if n is even,
we assume that $L > 0$.

Two-Path Test for Nonexistence of a Limit

If a function $f(x, y)$ has different limits along two different paths in the domain of f as (x, y) approaches (x_0, y_0) , then $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ does not exist.

Continuity:

DEFINITION A function $f(x, y)$ is **continuous at the point** (x_0, y_0) if

1. f is defined at (x_0, y_0) ,
2. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ exists,
3. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$.

A function is **continuous** if it is continuous at every point of its domain.

Partial Derivatives

DEFINITION The partial derivative of $f(x, y)$ with respect to x at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the limit exists.

$$f_x = \frac{\partial f}{\partial x} = \frac{\Delta f}{\Delta x}$$

$$f_y = \frac{\partial f}{\partial y} = \frac{\Delta f}{\Delta y}$$

Partial Derivates: $f(x, y)$

$$\bullet \quad f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}$$

• Finding derivatives w.r.t. to one variable at a time, treating all other elements as constants.

$$\bullet \quad f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), \quad f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right), \quad f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right).$$

Chain Rule: $f(t) = f(x(t), y(t))$.

$$\bullet \quad \frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Recall: from singular-variable calculus:
If $f(t) = f(x(t))$.

$$\frac{df}{dt} = \frac{df}{dx} \frac{dx}{dt}$$

Chain Rule:

Last time: $f(x, y), x(t), y(t) \rightarrow f(x(t), y(t))$

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

What if we have two independent variables for x, y , and z ?

$$w(x, y, z), x(r, \theta), y(r, \theta), z(r, \theta) \rightarrow w(x(r, \theta), y(r, \theta), z(r, \theta))$$

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r}.$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial \theta}.$$

Gradient

↗ this is a vector

$$\nabla f(x_0, y_0, z_0)$$

← at a point

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

$$\nabla f(x_0, y_0, z_0) = \left\langle \frac{\partial f}{\partial x}(x_0, y_0, z_0), \frac{\partial f}{\partial y}(x_0, y_0, z_0), \frac{\partial f}{\partial z}(x_0, y_0, z_0) \right\rangle$$

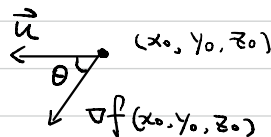
Directional Derivative

$$D_{\vec{u}} f(x_0, y_0, z_0)$$

↖ unit vector
(direction)

← at a point.

$$\begin{aligned} D_{\vec{u}} f(x_0, y_0, z_0) &= \nabla f(x_0, y_0, z_0) \cdot \vec{u} \quad \text{dot product} \\ &= |\nabla f(x_0, y_0, z_0)| |\vec{u}| \cos(\theta) \quad \text{unit vector.} \end{aligned}$$



Observations (Properties):

- ① when $\vec{u}$ and ∇f are at the same direction, f increases most rapidly
b/c $\theta = 0 \Rightarrow \cos(\theta) = 1$ (which is max of $\cos(\theta)$).
- ② when $\vec{u}$ is at the opposite direction as ∇f , f decreases most rapidly
b/c $\theta = \pi \Rightarrow \cos(\theta) = -1$ (which is min of $\cos(\theta)$).
- ③ when $\vec{u}$ is orthogonal to ∇f , f doesn't change
b/c $\theta = \frac{\pi}{2} \Rightarrow \cos(\theta) = 0 \Rightarrow D_{\vec{u}} f = |\nabla f| |\vec{u}| \cos(\theta) = 0$.

Tangent Line to a Level Curve @ (x_0, y_0)

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0$$

The tangent line to a level curve at point (x_0, y_0) is always orthogonal to the gradient at this point.

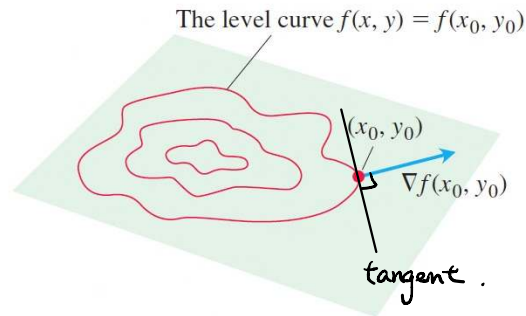


FIGURE 14.31 The gradient of a differentiable function of two variables at a point is always normal to the function's level curve through that point.

Tangent Plane & Normal Line

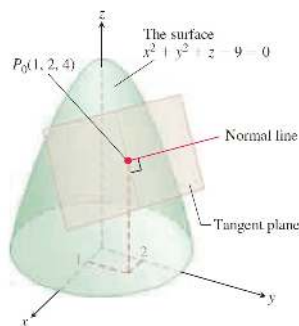


FIGURE 14.34 The tangent plane and normal line to this level surface at P_0 (Example 1).

Case 1: Tangent plane to the level surface $f(x, y, z) = c$

Tangent Plane to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0 \quad (1)$$

Normal Line to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$x = x_0 + f_x(P_0)t, \quad y = y_0 + f_y(P_0)t, \quad z = z_0 + f_z(P_0)t \quad (2)$$

Direction of the normal line: $\nabla f(x_0, y_0, z_0)$

Case 2: Tangent plane to a surface $f(x, y)$

Plane Tangent to a Surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$

The plane tangent to the surface $z = f(x, y)$ of a differentiable function f at the point $P_0(x_0, y_0, z_0) = (x_0, y_0, f(x_0, y_0))$ is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0. \quad (3)$$

Linearization & Total Differential

Ideal: Use a linear function to approximate the function value around a point (x_0, y_0, z_0) .

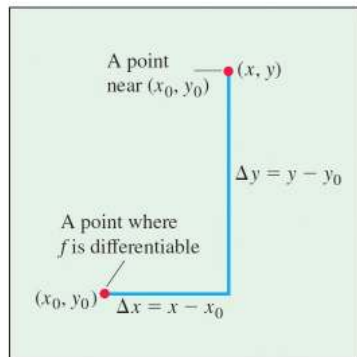


FIGURE 14.37 If f is differentiable at (x_0, y_0) , then the value of f at any point (x, y) nearby is approximately $f(x_0, y_0) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$.

DEFINITIONS The **linearization** of a function $f(x, y)$ at a point (x_0, y_0) where f is differentiable is the function

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The approximation

$$f(x, y) \approx L(x, y)$$

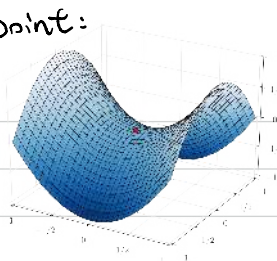
is the **standard linear approximation** of f at (x_0, y_0) .

DEFINITION If we move from (x_0, y_0) to a point $(x_0 + dx, y_0 + dy)$ nearby, the resulting change

$$df = f_x(x_0, y_0) dx + f_y(x_0, y_0) dy$$

in the linearization of f is called the **total differential of f** .

A saddle point:



Optimization

Summary of Max-Min Tests

The extreme values of $f(x, y)$ can occur only at

- i) **boundary points** of the domain of f
- ii) **critical points** (interior points where $f_x = f_y = 0$ or points where f_x or f_y fails to exist)

If the first- and second-order partial derivatives of f are continuous throughout a disk centered at a point (a, b) and $f_x(a, b) = f_y(a, b) = 0$, the nature of $f(a, b)$ can be tested with the **Second Derivative Test**:

- i) $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at $(a, b) \Rightarrow$ **local maximum**
- ii) $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at $(a, b) \Rightarrow$ **local minimum**
- iii) $f_{xx}f_{yy} - f_{xy}^2 < 0$ at $(a, b) \Rightarrow$ **saddle point**
- iv) $f_{xx}f_{yy} - f_{xy}^2 = 0$ at $(a, b) \Rightarrow$ **test is inconclusive**

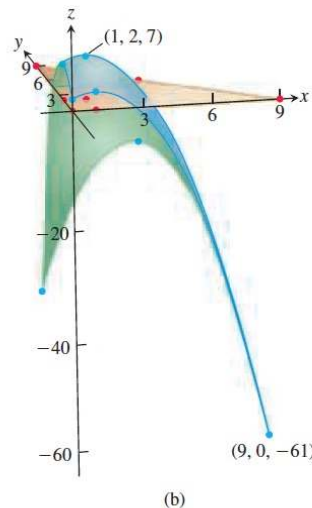
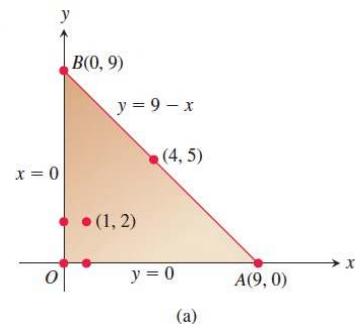


FIGURE 14.50 (a) This triangular region is the domain of the function in Example 6. (b) The graph of the function in Example 6. The blue points are the candidates for maxima or minima.

Constrained Optimization (Lagrange Multiplier)

- ① Identify the objective function $f(x, y, z)$ and the constraints $g(x, y, z) = C \leftarrow \text{constant}$.
- ② Find ∇f and ∇g
- ③ Using $\nabla f = \lambda \nabla g$ and $g(x, y, z) = C$ to set up equations

$$\begin{cases} f_x(x, y, z) = \lambda g_x(x, y, z) \\ f_y(x, y, z) = \lambda g_y(x, y, z) \\ f_z(x, y, z) = \lambda g_z(x, y, z) \\ g(x, y, z) = C \end{cases} \Rightarrow \text{Solve for candidate points}$$

Question: if a optimization have an objective function $f(x, y, z, w, u)$ with 3 constraints, how many equations should we set up?
($g(x, \dots, u) = a$, $h(x, \dots, u) = b$, $l(x, \dots, u) = c$)

Ans: 8 equations (5 from $\nabla f = \lambda_1 \nabla g + \lambda_2 \nabla h + \lambda_3 \nabla l$ and 3 from constraints)